

Insulin-like growth factor (IGF)-I, IGF binding protein-3, and cancer risk: systematic review and meta-regression analysis.

BACKGROUND: Insulin-like growth factor (IGF)-I and its main binding protein, IGFBP-3, modulate cell growth and survival, and are thought to be important in tumour development. Circulating concentrations of IGF-I might be associated with an increased risk of cancer, whereas IGFBP-3 concentrations could be associated with a decreased cancer risk. **METHODS:** We did a systematic review and meta-regression analysis of case-control studies, including studies nested in cohorts, of the association between concentrations of IGF-I and IGFBP-3 and prostate, colorectal, premenopausal and postmenopausal breast, and lung cancer. Study-specific dose-response slopes were obtained by relating the natural log of odds ratios for different exposure levels to blood concentrations normalised to a percentile scale. **FINDINGS:** We identified 21 eligible studies (26 datasets), which included 3609 cases and 7137 controls. High concentrations of IGF-I were associated with an increased risk of prostate cancer (odds ratio comparing 75th with 25th percentile 1.49, 95% CI 1.14-1.95) and premenopausal breast cancer (1.65, 1.26-2.08) and high concentrations of IGFBP-3 were associated with increased risk of premenopausal breast cancer (1.51, 1.01-2.27). Associations were larger in assessments of plasma samples than in serum samples, and in standard case-control studies compared with nested studies. **INTERPRETATION:** Circulating concentrations of IGF-I and IGFBP-3 are associated with an increased risk of common cancers, but associations are modest and vary between sites. Although laboratory methods need to be standardised, these epidemiological observations could have major implications for assessment of risk and prevention of cancer.